

MD RAKIBUL ISLAM RAIHAN

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Nationality: Bangladesh

MBTI: ENTJ

Languages: English (Fluent, IELTS 8.0), Chinese (Conversational, HSK 3)

LinkedIn: <https://www.linkedin.com/in/theraihanrakib>

Portfolio: <https://theraihanrakibb.github.io/Online-Portfolio/>

Seeking Positions: AI Algorithm R&D Engineer (Multimodal / AIGC / LLM-VLM) | AI Product Manager



Objective

Master's candidate in Software Engineering (**top 1% of program**) with current industry experience designing **NVFP4 pretraining pipelines for LLMs and VLMs**, optimizing inference serving (**SGLang, vLLM, TensorRT-LLM, Transformer Engine**), and developing **custom CUDA/Triton kernels** for mixed-precision training.

Research focuses on **multimodal AI, generative models (Diffusion, GANs)**, and **audio-visual fusion** — from fundamentals to deployment. Proficient in **PyTorch, TensorFlow**, and **Apache Spark** for large-scale data processing. Previously led an **end-to-end text-to-video generation pipeline** in an academic lab and delivered **full-stack AI Agent solutions** with commercial deployment.

Committed to advancing foundation models and multimodal algorithms. Seeking **AI Algorithm R&D Intern** (Multimodal / AIGC / Computer Vision / Robotics) or **AI Product Manager Intern** role bridging cutting-edge research and real-world products to drive impactful innovation.

Education

Master of Engineering in Software Engineering

Northwestern Polytechnical University, Xi'an, China (**QS #499, 985, 211, E9 League**)

Sep 2024 – Mar 2027

- **GPA:** 88/100
- **Rank:** 1%
- **Relevant Coursework:** Machine Learning, Computer Vision, Natural Language Processing, Software Project Management.
- **Extracurricular Activities:** Member of Student Union, Academic Mentor, Teaching Assistant.
- **Research:** "Multimodal Video Generation: Integrating Audio and Visual Signals for Realistic Content Creation" (Under the guidance of Dr. Zhang Peng).
- **Thesis:** "Detecting Deepfake Videos: A Multimodal AI Framework Leveraging Visual, Auditory, and Temporal Inconsistencies." (Under the guidance of Dr. Zhang Peng).

Bachelor of Engineering in Computer Science and Technology

Northwestern Polytechnical University, Xi'an, China (**QS #499, 985, 211, E9 League**)

Sep 2020 – Jul 2024

- **GPA:** 85/100
- **Rank:** 1%

- **Outstanding Undergraduate Graduate (Top 10)** ([link](#))
 - **Relevant coursework:** Algorithms, Data Structures, Software Engineering, Database Systems.
 - **Extracurricular Activities:** Director of International Student Union (2023), Academic Volunteer (2023,2024)
 - **Thesis:** “Design and implemented a Distributed Confidential Query Protocol for Apache Spark.” (Under the guidance of Dr. Chen Yaxing).
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Experience

Graduate Research Assistant

Multimodal Learning and Video Generation Lab, NWPU

Sep 2024 – Mar 2027

Advisor: Dr. Zhang Peng

- Spearheaded research in **audio-visual fusion models** to enhance the realism and synchronization of generated video content.
- Designed and implemented a novel **self-supervised learning framework** for cross-modal representation learning, improving alignment between audio and visual data streams.
- Engineered an end-to-end **text-to-video generation pipeline** leveraging Transformer-based architectures (e.g., Diffusion Transformers) to convert semantic prompts into dynamic visual sequences.

Undergraduate Research Intern

High Performance Computing Lab, NWPU

Jan 2023 – Jul 2024

Advisor: Dr. Chen Yaxing

- Researched and implemented a **Distributed Confidential Query Protocol**, designing mechanisms for secure, privacy-preserving data retrieval across distributed nodes.
 - Developed large-scale data processing modules for a **cloud computing and networking project** using **Apache Spark and SQL**, optimizing query performance on distributed datasets.
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Internship

Technical Staff Intern – AI Infra

[InfiX.ai](#) LIMITED

Apr 2026 – Oct 2026

- Design and implement NVFP4 pretraining pipelines for LLMs and VLMs, targeting 4-bit low-precision training at scale.
- Optimize inference serving using SGLang, vLLM, TensorRT-LLM, and Transformer Engine with NVFP4 tensor cores to improve throughput and latency.
- Develop custom CUDA/Triton kernels for packed FP4 data formats and mixed-precision logic, enabling efficient memory access and compute.
- Profile and debug distributed training and inference workloads using Nsight Systems and Nsight Compute to ensure full hardware utilization.
- Validate convergence and model quality of FP4 training against BF16 baselines across multiple architectures and tasks.
- Collaborate on quantization-aware training (QAT) and block-wise scaling strategies to narrow the accuracy gap at extreme low precision.

Software Engineer – AI Agent

[Hong Kong Amraim Group Co., Ltd.](#)

Jun 2025 – Sep 2025

- Engineered intelligent AI agents that automated key marketing workflows, reducing manual effort and boosting team productivity by 25%.

- Architected and delivered two end-to-end full-stack applications from concept to deployment, directly addressing core business requirements and user needs.
- Built dynamic, responsive front-end interfaces using React and developed scalable back-end services with RESTful APIs to seamlessly integrate AI/ML features.
- Drove the full software development lifecycle in an Agile (Scrum) environment, from planning to deployment, ensuring the delivery of secure, scalable, and production-ready solutions.

Electrical Software Engineer – Algorithms

[Shaanxi Longong Intelligent Technology Co., Ltd.](#)

Feb 2025 – Apr 2025

- Engineered and optimized real-time motor control algorithms for industrial automation machinery, improving system response time by 15% and increasing overall throughput.
- Developed, debugged, and validated embedded control software utilizing PLC programming (Codesys) and industrial communication protocols (Modbus, CAN, Ethernet/IP) to ensure reliable system operation.
- Designed electrical schematics and led prototyping for hardware-software interfaces in new automation equipment, ensuring full compliance with industry safety and regulatory standards (e.g., IEC 61131).
- Collaborated within a cross-functional R&D team using Agile (Scrum) methodologies; managed project tasks in JIRA and maintained version control with Git to streamline the development lifecycle.

Professional & Technical Skills

✧ AI/ML & Deep Learning

PyTorch • TensorFlow • Keras • Transformer Architectures • Self-Supervised & Multimodal Learning • Audio-Visual Fusion • Generative Models (Diffusion) • NVFP4 Low-Precision Training • vLLM • TensorRT-LLM • Transformer Engine • Quantization-Aware Training (QAT) • Block-Wise Scaling

✧ Systems & High-Performance Computing

CUDA • SGLang • Triton (Kernel Development) • NVIDIA Nsight Systems • NVIDIA Nsight Compute • Mixed-Precision Training • Distributed Model Fusion • FP8/FP4 Optimization

✧ Programming & Data Engineering

Python (Proficient) • C++ • Java • SQL • MATLAB • Apache Spark • Hadoop • Hive • RESTful APIs

✧ Cloud, MLOps & Embedded Systems

AWS • Google Cloud Platform (GCP) • Docker • Git • Embedded Systems (PLC, Codesys, Modbus/CAN/Ethernet/IP)

✧ Methodologies & Core Competencies

Agile/Scrum Project Management • Cross-Functional Collaboration • Full Software/Research Lifecycle • Technical Research & Writing • Complex Problem-Solving • Electrical Prototyping & Safety Compliance • Performance Profiling & Debugging

Projects

1. Alcohol Detection System for Automobiles | *Embedded IoT Project*

- Engineered a real-time alcohol detection prototype for automotive safety using an IR sensor and an Arduino/ESP32 microcontroller.
- Developed the sensor data processing firmware and a threshold-based alert system to trigger warnings upon detection.
- Designed the hardware interface and enclosure, ensuring reliability for in-vehicle environmental conditions.

2. Distributed Confidential Query Protocol for Apache Spark | *Privacy-Preserving Big Data*

- Designed and implemented a secure, privacy-preserving query execution engine for Apache Spark using homomorphic encryption.
- Maintained 95% data confidentiality with minimal performance overhead during testing on a 10-node cluster.

- Optimized distributed query performance by 20% compared to baseline privacy frameworks.

3. Multimodal Text-to-Video Generation Tool | *Generative AI Research*

- Built an end-to-end text-to-video generation pipeline using Diffusion Transformer (DiT) architectures and GAN components.
- Integrated cross-modal (audio-visual) alignment techniques, with generated outputs achieving 90% user satisfaction in qualitative evaluations.
- Deployed the scalable inference service on AWS EC2 & S3, enabling on-demand video synthesis.

4. GAN-Based Medical Image Enhancement | *Healthcare AI Application*

- Developed a conditional GAN model to super-resolve and denoise low-resolution MRI scans, enhancing clinical detail.
- The enhanced images improved downstream tumor detection accuracy to 85% when validated with a standard segmentation model.

5. Real-Time Visual Object Tracking for UAVs | *Computer Vision & Autonomous Systems*

- Designed and trained a deep learning-based visual tracking system for unmanned aerial vehicles (UAVs) in dynamic environments.
- Achieved 95% tracking accuracy (IoU) when evaluated on the BAIR Robot Pushing Dataset, demonstrating robustness to occlusion.

Certifications

- **AWS Certified Machine Learning – Specialty | Amazon Web Services**
(Validates ability to ship production ML solutions on a major cloud platform)
- **Deep Learning Specialization | [DeepLearning.AI](#) (Coursera)**
(Certifies core AI/ML knowledge for informed technical decision-making and roadmap planning)
- **Agile & Product Methodology**
Professional Scrum Master I (PSM I) | [Scrum.org](#)
(Demonstrates formal Agile/Scrum expertise for leading technical teams and managing development cycles)

Awards and Achievements

Scholarships & Academic Honors

- Chinese Government Scholarship (2024 – 2026 | Awarded for academic excellence)
- NWPU Presidential Scholarship (2020 – 2023 | Awarded consecutively for top-tier academic performance)
- Sanqin Scholarship (2023 | Merit-based award for outstanding research potential)
- Wu Yajun Scholarship (2024 | Merit-based award for outstanding research potential)

Leadership & Service

- Director, International Students Union (2023 | Led initiatives and represented a diverse student body)
- Academic Volunteer (2022 – 2023 | Assisted in tutoring and research support for underclassmen)

Competitions & Recognition

- Hackathon Winner, NWPU AI Challenge (2022 | Led team to victory in 48-hour applied AI competition)
- Best Undergraduate Thesis Award – *Distributed Confidential Query Protocol for Spark* (Awarded for top research contribution and implementation)